A Study on Conductor Loss Measurement of Microstrip Line

Yusuke Kusama, and Yusuke Yokoi
Dept. of communication and network engineering
National Institute Technology, Kagawa College
551 Kouda, Takuma-cho, Mitoyo City, Kagawa, Japan
kusama@cn.kagawa-nct.ac.jp

Robert Weston Johnston
Dept. of electronic system engineering
National Institute Technology, Kagawa College
551 Kouda, Takuma-cho, Mitoyo City, Kagawa, Japan

***Abstract*—Many studies on the conductor loss of the copper-clad dielectric substrate have already been reported. And a method of obtaining frequency characteristics of the conductivity from the measured value by applying the concept of effective conductivity is adopted. In this report, the decrease rate of the signal line cross-sectional area due to the skin effect is estimated from the transmission characteristic measurement according to the physical law that the cross-sectional area of the signal line decreases due to the skin effect instead of the conductivity frequency characteristic.**

***Keywords*—microwave circuit; microstrip ; conductor loss; skin effect; education**

I. INTRODUCTION

It has been said human resources in microwave circuits and antenna designs having expertise and practical skills are in a state of chronic shortage in Japan [1]. In order to train excellent microwave engineers, it is essential to experience theoretical calculations and experiments with the help of electromagnetic simulators to deepen both knowledge and practical skills [2]. From the viewpoint of student educations, lectures are the basic acquisition of the theoretical knowledge, but in microwave engineering, it includes applied mathematics, programming, simulations required for analyzing circuits, modules, materials, antennas, measurement techniques, error analysis, and theoretical analysis. In order to broaden the frontage for students who will become first scholars, it is necessary to repeat the systematic arrangement and maintenance of educational contents in accordance with the progress of the times and technologies. This requires extensive labors and costs, but acquisition forms of the theoretical knowledge are diversifying such as reverse lectures using a web such as Open Course Ware (OCW) or remote lecture deliveries.

Meanwhile, acquisitions of measurement techniques and discussion methods require repeated experience of trial and error cycles of theories, trial manufactures, measurements, and simulations. Introduction training periods in recent companies tends to increase year by year, which is considered due to the fact that specialized education contents are fixed to a certain extent both in theory and experiment, whereas it is becoming impossible to keep up with changes in technology speed in the world. In order to cope with such a situation, it is necessary to introduce and develop practical and flexible specialized experimental programs continuously. Overseas, numerous packages related to microwave educations for universities have been developed [3], and in recent years not only experimental equipment but also teaching slides and measurement spread sheets are bundled [4][5].

In such a background, it is considered developing engineering studies designing waveguide circuit elements and microstrip line (MSL) circuits as a microwave experiment package program combining theoretical calculations, simulations and measurements in National Institute of Technology Kagawa College (NIT-KC) [6]-[8]. The usual MSL circuit uses a low loss substrate, and practical problems are almost none by evaluating the conductor loss within a negligible frequency range. However, in order to apply to nearly disposable student experiments, it is firstly required to be lower in cost than the performance. In addition, from the viewpoint of student experiments at the undergraduate level, it is also necessary to be able to discuss at the level of general textbooks, which are widely spread in the world. We have been studied experiments on MSL basic circuit designs using FR-4 substrate so far, but it was known that considering dielectric loss only greatly deviates from the measured values when it exceeds 6 GHz [6][7]. Therefore, the conductor loss, which has not been considered so far, has been necessary from the viewpoint of considerations in student experiments.

II. METHOD FOR EVALUATING CONDUCTOR LOSS

Many studies on the conductor loss of copper-clad dielectric substrates have already been reported [9][10], most of which introduces the concept of an effective conductivity and adopts the method of finding frequency characteristics of the conductivity from the measured value by using the accurate resonator method [11]-[21]. However, according to Ref. [22], the calculated frequency dependence of the electrical characteristic of copper is constant up to 4 THz of the relaxation frequency at which electrons start to invert their direction before colliding with the crystal atoms. Taking this into account, it is not realistic to consider that the physical conductivity of copper changes depending on the frequency in the microwave band of several GHz. Therefore, in this paper, in accordance with the physical law that the cross sectional area of the signal line decreases due to the skin effect, instead of the conductivity frequency characteristic itself, a method of estimating the reduction rate of the signal line cross sectional area from the transmission characteristic measurement value was studied. According to the premise that the attenuation constant α_d due to the dielectric loss is known, it is possible to estimate the attenuation constant α_c due to the conductor loss only from the transmission characteristic. However, the basic idea is the same as the concept of the effective conductivity proposed in the previous research since it is difficult to separate the conductivity σ and the cross-sectional area S as shown by the equation of resistance in equation (1).

$$R = \rho\frac{l}{S} = \frac{l}{\sigma S} \tag{1}$$

Table 1 shows the dielectric constant nominal values of the FR-4 substrate used in this experiment. It is a glass cloth based epoxy resin copper clad laminate NZ-30KR manufactured by Sunhayato Corporation. Although, high frequency characteristics are not described, the dielectric constant is almost constant $\varepsilon_r = 4.4$ and $\tan\delta = 0.02$ up to about 10 GHz from the previous study [6]. This value is the same as the FR4_epoxy which is incorporated in ANSYS HFSS material library. Note that we tried dielectric constant derivation by using MSL $\lambda/2$ open resonator and $\lambda/4$ open resonator [23], but error is as large as several tens% and good results are not obtained because of the influence of fringing effect.

TABLE I. NOMINAL VALUE OF SUBSTRATE

Material	**Glass epoxy (FR-4)**
Dielectric constant	4.5 - 4.9 @ 1 MHz
Dissipation factor	0.013 - 0.02 @1 MHz
Copper foil thickness	35 µm
Substrate thickness	1.6 mm
Water absorption rate	0.05 - 0.15%

III. CALCULATION

A. Dielectric loss calculation method

Table 2 shows the propagation constant α, β and characteristic impedance Z_0 when the transmission line is treated as a lumped constant circuit element and the propagation constant α, β and wave impedance Z_w when treated as the guided wave. Table 3 shows the similarity between the distributed constant line and the plane wave as an equivalent circuit. In order to evaluate the conductor loss by using the distributed constant line, it is necessary to derive the frequency characteristic of R of the unit length. Since the internal inductance L_e of the conductor gradually approaches zero by the skin effect, it can be considered that only the resistance component R needs to be considered. However, except for cases of axisymmetric closed spaces such as a coaxial line, its derivation is not easy [24]-[26]. On the other hand, the transmission power in the TEM mode is the same for both the distributed constant line and the plane wave. Although the main mode of the MSL is the quasi-TEM mode, if the effective permittivity ε_e is introduced instead of the substrate dielectric constant ε_r, the dielectric loss can also be represented by the attenuation constant of the plane wave using equation (2).

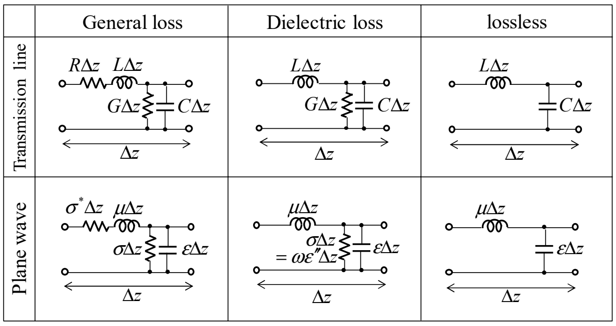
$$\alpha_d = \omega\sqrt{\frac{\mu\varepsilon_e}{2}}\left[\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}-1\right]^{\frac{1}{2}} \tag{2}$$

where σ is the loss of the substrate, not the conductivity of the conductor. And $\sigma/\omega\varepsilon$ is equals to $\tan\delta$.

TABLE II. PROPAGATION CONSTANTS

		General Loss	Dielectric Loss
Transmission Line (V, I)	α	$\sqrt{\frac{\omega^2 LC-RG}{2}}\left[\sqrt{1+\omega^2\left(\frac{CR+LG}{\omega^2 LC-RG}\right)^2}-1\right]^{\frac{1}{2}}$	$\omega\sqrt{\frac{LC}{2}}\left[\sqrt{1+\left(\frac{G}{\omega C}\right)^2}-1\right]^{\frac{1}{2}}$
	β	$\sqrt{\frac{\omega^2 LC-RG}{2}}\left[\sqrt{1+\omega^2\left(\frac{CR+LG}{\omega^2 LC-RG}\right)^2}+1\right]^{\frac{1}{2}}$	$\omega\sqrt{\frac{LC}{2}}\left[\sqrt{1+\left(\frac{G}{\omega C}\right)^2}+1\right]^{\frac{1}{2}}$
	Z_0	$\frac{R+j\omega L}{\gamma}=\sqrt{\frac{R+j\omega L}{G+j\omega C}}$	$\sqrt{\frac{j\omega L}{G+j\omega C}}$
Plane Wave (E, H)	α	$\sqrt{\frac{\omega^2\mu\varepsilon-\sigma^*\sigma}{2}}\left[\sqrt{1+\omega^2\left(\frac{\varepsilon\sigma^*+\mu\sigma}{\omega^2\mu\varepsilon-\sigma^*\sigma}\right)^2}-1\right]^{\frac{1}{2}}$	$\omega\sqrt{\frac{\mu\varepsilon}{2}}\left[\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}-1\right]^{\frac{1}{2}}$
	β	$\sqrt{\frac{\omega^2\mu\varepsilon-\sigma^*\sigma}{2}}\left[\sqrt{1+\omega^2\left(\frac{\varepsilon\sigma^*+\mu\sigma}{\omega^2\mu\varepsilon-\sigma^*\sigma}\right)^2}+1\right]^{\frac{1}{2}}$	$\omega\sqrt{\frac{\mu\varepsilon}{2}}\left[\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}+1\right]^{\frac{1}{2}}$
	Z_w	$\frac{\sigma^*+j\omega\mu}{\gamma}=\sqrt{\frac{\sigma^*+j\omega\mu}{\sigma+j\omega\varepsilon}}$	$\sqrt{\frac{j\omega\mu}{\sigma+j\omega\varepsilon}}$

TABLE III. EQUIVALENT CIRCUIT



As an example, Fig. 1 shows $|S_{21}|$ characteristic $\lambda/4$ short-circuited resonator when conductor loss is ignored. The solid line shows HFSS with no internal analysis of conductors, the dotted line shows the calculation result using the propagation constant of the plane wave dielectric loss medium in Table 2, and the broken line shows the calculation result using the dielectric loss equation (3) in Ref. [27]. To consider the influence of dielectric losses, it suffices to multiply the lossless $|S_{21}|$ by the attenuation coefficient $\exp[-2\alpha_d\,(a + l)]$. Here, α_d is the damping constant, a is the length of arms, and l is the stub length. From this result, it can be seen that the dielectric loss can be evaluated by using the attenuation constant of the plane wave dielectric loss medium of Table 2.

$$\alpha_d = \frac{k_0\varepsilon_r\left(\varepsilon_e-1\right)\tan\delta}{2\sqrt{\varepsilon_e}\left(\varepsilon_r-1\right)} \tag{3}$$

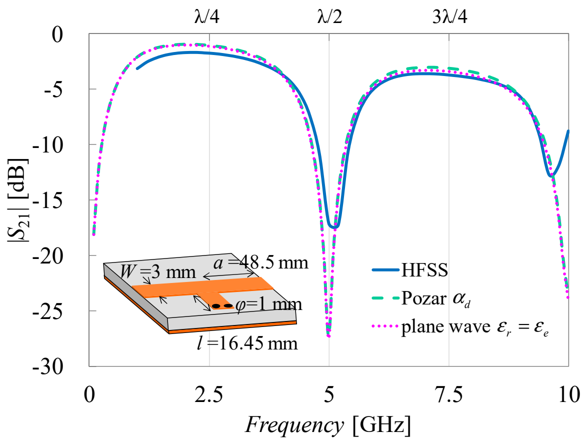


Fig. 1. Comparison of dielectric loss calculation formulas.

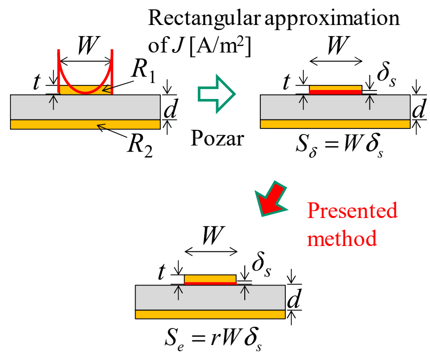


Fig. 2. Rectangular approximation of signal line current.

B. Method of calculation of conductor loss

In the cross-sectional model shown in Fig. 2, relational expressions between the conductor attenuation constant α_c and the resistance R per unit length are as follows. When the current distribution of the signal line is approximated by a rectangle $S_\delta = W\delta_s$ of the back surface skin depth S_δ as shown in the right side of Fig. 2, the resistance R_1 of the signal line per 1 m is

$$R_1 = \rho \frac{l}{S} = \rho \frac{1}{W\delta_S} \ [\Omega/\mathrm{m}] \tag{4}$$

where $\rho = 1/\sigma$ indicates the resistivity of copper, and S_δ is the skin depth given by the following equation (5).

$$\delta_S = \sqrt{\frac{2}{\omega\mu\sigma}} \tag{5}$$

Assigning equation (5) to (4) and rearranging is

$$R_1 = \frac{1}{W}\sqrt{\frac{\omega\mu}{2\sigma}} = \frac{R_S}{W} \tag{6}$$

where R_s is the surface resistivity of the conductor. For simplicity, if the resistor R_2 of GND is equal to the resistance R_1 of the signal line ($R_1 \simeq R_2$), the resistance R per unit length when approximating the current rectangle by the skin depth and the line width is

$$R = R_1 + R_2 = \frac{2R_S}{W} \ [\Omega/\mathrm{m}] \tag{7}$$

Conductor loss attenuation constant α_c per unit length is

$$-\alpha_c = \frac{1}{2}\ln\frac{P_{\mathrm{out}}}{P_{\mathrm{in}}} = \frac{1}{2}\ln\frac{P_{\mathrm{in}} - P_{\mathrm{Loss}}}{P_{\mathrm{in}}} \ [\mathrm{Np/m}] \tag{8}$$

Here, when the following equation is used for the resistance loss power P_{Loss} and $I^2 = P/Z_0$

$$P_{\mathrm{Loss}} = P_{\mathrm{in}} - P_{\mathrm{out}} = RI^2 = R\frac{P_{\mathrm{in}}}{Z_0} \tag{9}$$

Assigning equation (9) into (8) and rearranging is

$$\alpha_c = -\frac{1}{2}\ln\left(1 - \frac{R}{Z_0}\right) \tag{10}$$

When $R \ll Z_0$ (condition of good conductor), equation (10) can be approximated by the following equation (11).

$$\alpha_c = \frac{1}{2}\frac{R}{Z_0} \tag{11}$$

Also, substituting equation (7) into equation (11)

$$\alpha_c = \frac{R_s}{Z_0 W}, \quad R_s = \sqrt{\frac{\omega\mu_0}{2\sigma}} \tag{12}$$

is obtained. Equation (12) is an approximate expression of the conductor loss described in Ref. [27]. In both equations (11) and (12), the resistance value increases due to the effect of the skin effect, and caution is required when applying when the condition of good conductor is not satisfied.

It is known that conductor loss hardly occurs even when Eq. (12) is applied to the FR-4 substrate, and it deviates greatly from the measured value from 6 GHz [6]. This is considered that the skin depth S_δ calculated by the equation (5) becomes 1 µm or less when it exceeds 4 GHz, but the signal line width W is 3,000 times that, so the rectangular cross-sectional area S_δ is too large in the direction of the line width. Therefore, in this study, the attenuation constant α_c was determined so as to fit to the measured $|S_{21}|$ to approximate the actual sectional current distribution. Specifically, we set the effective conductor cross-section decreasing due to the skin effect as $S_e = rS_\delta$ ($r \leq 1$), and decide the coefficient r so that it coincides with measured $|S_{21}|$. At this time, the effective resistance R_e is expressed by the following equation (13).

$$R_e = \frac{2}{\sigma S_e} = \frac{2}{\sigma r S_\delta} = \frac{2}{r\sigma W S_\delta} \tag{13}$$

Although the attenuation constant α_c is represented strictly by the equation (10), in the case of the condition $R \gg Z_0$ where the effective conductor cross-sectional area S_e decreases due to the skin effect and can not be regarded as a good conductor, it is impossible to directly obtain α_c from Eq. (10). Therefore, α_c is also derived approximately by using Equation (11) which is the first term when Taylor's expansion of Equation (10) is approximated.

IV. MEASUREMENT

A. *Measurement sample*

Figure 3 shows the schematic view of the measurement sample used in the experiment. The samples are lines, $\lambda/4$ short-circuit resonators, and $\lambda/4$ open resonators. Signal line patterns are fabricated by etching with a line width W = 3 mm so that Z_0 = 50 Ω. The resonator dimension l is determined to resonate at 2.5 GHz. Other dimensions are shown in the figure. Since etching conditions are based on the manual [28] prepared for student experiments, details are omitted.

B. *Measurement system*

Fig. 4 shows the measurement system. 3.5 mm coaxial connector tip was set on the calibration surface, then measurement samples were inserted and the transmission characteristic S_{21} was measured in the range of 0.01 - 10 GHz. The instrument used was a vector network analyzer Agilent E8362C with an IF band of 1 kHz and measuring points of 401 points. In addition, SMA connectors RS 526-5785 were attached to both ends of the substrate for connection with the coaxial calibration surface. As a measurement representative value, one having relatively stable characteristics from among four or more samples was adopted.

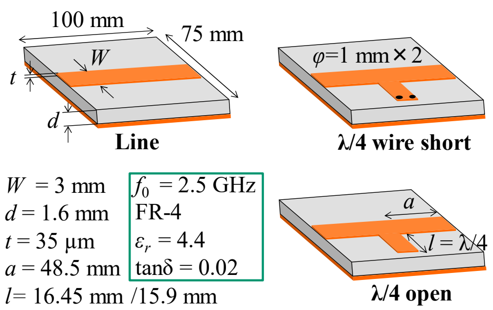


Fig. 3. Measurement samples.

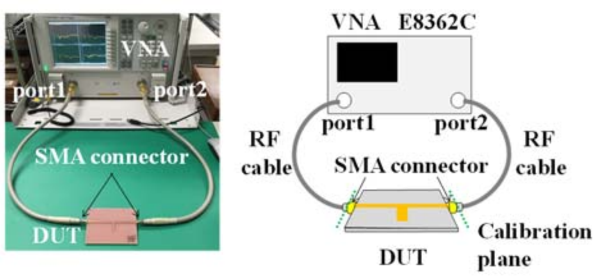


Fig. 4. Schematic view of measurement system.

V. RESULTS

Fig. 5 shows the transmission characteristics of the line. The four kinds of data show measurements fitted using measured values, HFSS, calculated values (hereinafter referred to as Pozar) considering both dielectric loss of equation (3) and conductor loss of equation (12). From this result, it is confirmed that the characteristics of HFSS with no inner conductor analysis and Pozar are almost identical.

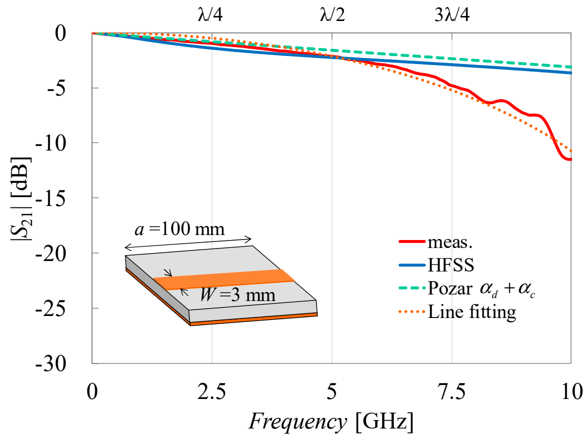


Fig. 5. Measurement result of line.

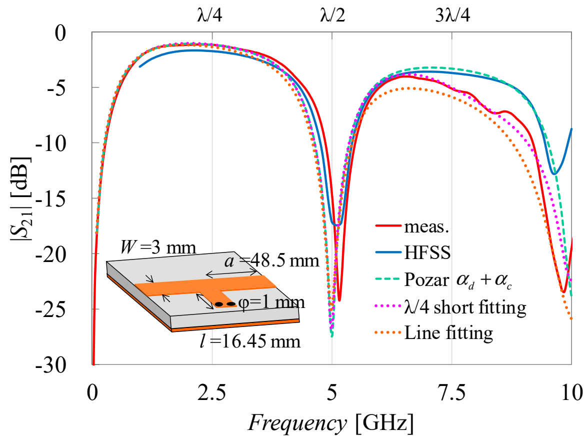


Fig. 6. Measurement result of $\lambda/4$ short resonator.

On the other hand, it is observed that the attenuation of measured value increases from around 6 GHz, and conductor loss exceeds the dielectric loss at 10 GHz. In addition, Fig. 6 also shows the transmission characteristics of $\lambda/4$ short-circuited stub. Five kinds of data show measured value, HFSS, Pozar, and two types of fittings to measured values. Two types of fittings show applying α_c obtained by fitting to the measured $|S_{21}|$ of the $\lambda/4$ short-circuited resonator, and fitting to the measured $|S_{21}|$ of the line of Fig. 5. When α_c of the line is applied, the attenuation is excessive by 2 dB over the measured value. The reason is considered that lossless $|S_{21}|$ is multiplied by exp[-2 $(\alpha_c + \alpha_d)(a + l)$] as the attenuation amount calculation, but the resonator length l is slightly shortened by the short-circuited wire, and T-junction shortening by the capacitive coupling [29][30]. In addition, Fig. 7 shows the frequency characteristics of the cross-sectional reduction ratio r in equation (13). The solid line shows the value determined to be fitted manually to measured $|S_{21}|$, and the dotted line shows the case of the function fitting. The function used for the line is shown in equation (14), and for the $\lambda/4$ short-circuited resonator is shown in equation (15).

$$r_{\mathrm{Line}} = \sqrt{\frac{a}{\omega \times 10^{-9}}} - b \qquad (14)$$

where $a = 6.3$, $b = 0.3$.

$$r_{\lambda/4} = \frac{1-b}{1+\exp\left[a\left(\omega\times 10^{-9} - c\right)\right]} + b \qquad (15)$$

where $a = 0.3$, $b = 0.05$, $c = 37.7$.

Since the influence of dielectric loss is larger than conductor loss up to 6 GHz, the difference between the line and $\lambda/4$ short does not matter. However when it exceeds 7 GHz, it can be observed that they agree with each other in 5% or less of the rectangular approximate cross section S_δ. Next, the effective resistance derived from equation (13) is shown in Fig. 8. The effective resistance per unit length rapidly increases from around 5 GHz, and becomes larger than the characteristic impedance 50 Ω at 6 GHz. Furthermore, when it exceeds 9 GHz, it becomes about 10 times of Z_0. Further, Fig. 9 shows the frequency characteristics of the attenuation constant α_c derived from the conductor losses derived from equations (10), (11) and (12). For comparison, the attenuation constant due to the dielectric loss derived from equations (2) and (3) are shown. It was found that attenuation due to conductor loss becomes dominant both line and $\lambda/4$ short when exceeded 7 GHz. The fact that R becomes too large to be ignored means that the characteristic impedance Z_0 becomes a complex number. Then it will be difficult to achieve matching at a higher frequency. Finally, in order to confirm the validity of proposed method, the case where the attenuation constant due to the conductor loss obtained from the line, and $\lambda/4$ short circuit resonator are applied to the $\lambda/4$ open resonator. The results are shown in Fig. 10. The five kinds of data are the same as those in the case of the short-circuited resonator in Fig. 6. It was confirmed that by using α_c obtained from the line fitting, the attenuation characteristics agree well up to about 10 GHz with considering the fringing effect.

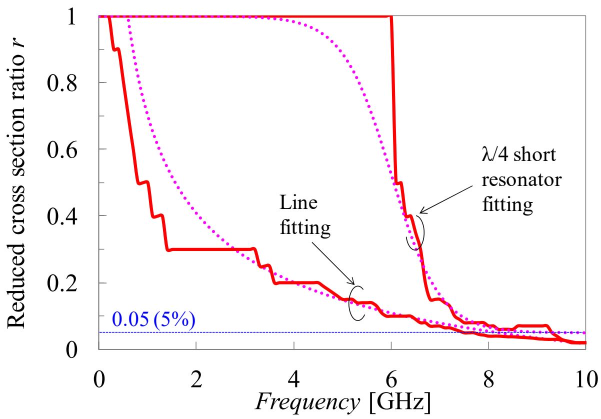


Fig. 7. Frequency characteristics of section reduction rate r.

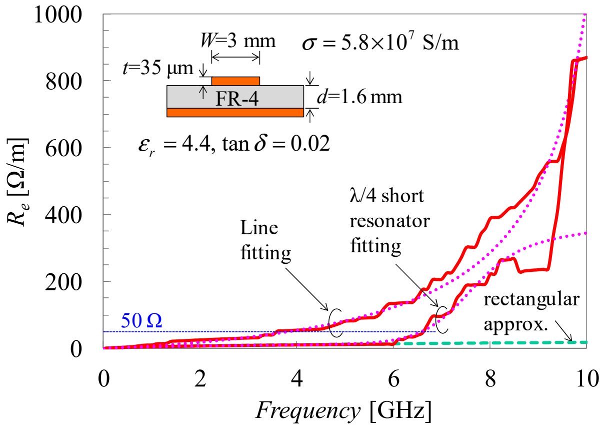


Fig. 8. Estimated resistance.

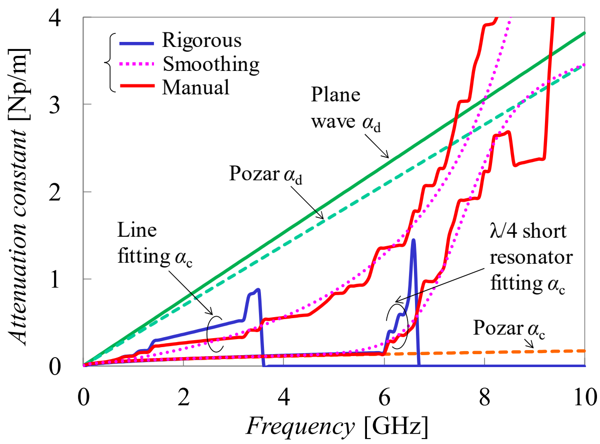


Fig. 9. Estimated attenuation constant.

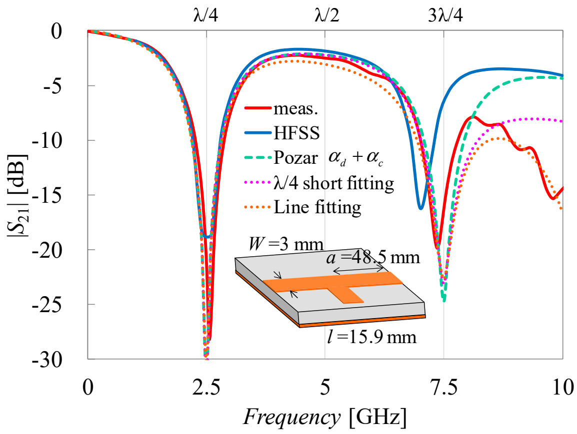


Fig. 10. Measurement result of $\lambda/4$ open resonator. (Measurement value and HFSS only have fringe effect correction, others are $l = 16.45$ mm).

VI. Conclusion

As a part of the development of practical program for microwave engineer training, we have been studied the development of experiment themes to acquire a series of trial and error processes ranging from designs, trial manufactures, simulations, measurements and evaluations. As one of them, we have been studied the theme of basic

MSL circuit fabrication, but when using low cost FR-4 substrate suitable for student experiment close to disposable, it was an issue that the effect of conductor loss could not be ignored in the frequency above 6 GHz. Although, it is same as the concept of an effective conductivity already reported, the decrease rate of the signal line cross-sectional area due to the skin effect was estimated from the transmission characteristic measurements under the premise of the dielectric loss is known. The attenuation characteristic of $\lambda/4$ short and open resonator were in good agreement with the measured value applying the attenuation constant due to the conductor loss obtained from the line measured value. From the above, the validity of this method could be confirmed. In the future, we plan to introduce and develop practical and flexible specialized experiment program by increasing variations of MSL circuit pattern.

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